

**BCE-P517 / P730**

**SEMESTER EXAMINATION-2022**

**B.TECH. – V/VII SEMESTER**

**Paper Title : MACHINE LEARNING-I**

206301076

**Time: 3 hours**

**Max. Marks: 70**

**Note:** Question Paper is divided into two sections: A and B. Attempt both the sections as per given instructions.

**SECTION-A (SHORT ANSWER TYPE QUESTIONS)**

**Instructions:** Answer any five questions in about 150 words each. Each question carries six marks.  
(5 × 6 = 30 Marks)

Question-1: Compare Batch/Offline and Online Learning with suitable example.

Question-2: Discuss random data sampling and stratified data sampling technique in brief.

Question-3: Explain mean, median and mode using appropriate example.

Question-4: Define machine learning. Also explain Artificial Narrow Intelligence (ANI), Artificial General Intelligence (AGI), Artificial Super Intelligence (ASI) in short.

Question-5: Explain data imbalance problem with respect to classification in brief. Also list different techniques to handle it.

Question-6: What is reinforcement learning? Compare Model-Free and Model-Based reinforcement learning in short.

Question-7: Discuss various applications and limitations of machine learning.

Question-8: Explain "Curse of Dimensionality" in reference to ML model complexity and performance.

Question-9: Differentiate structured and unstructured data with respect to ML.

Question-10: Discuss various important points to be kept in mind before and during installation of ANACONDA IDE if you are a ML developer working different type of ML Projects.

## SECTION-B (LONG ANSWER TYPE QUESTIONS)

**Instructions:** Answer any **four** questions in detail. Each question carries 10 marks.

(4 × 10 = 40 Marks)

Question-1: Explain Overfitting and Underfitting of ML Model in brief. Also discuss how linear regression model finds best fit line using gradient decent method using any example.

Question-2: Explain different types of machine learning using appropriate example. Also differentiate between supervised and unsupervised machine learning in brief.

Question-3: Draw neat and clean diagram of ML process flow and explain different steps involved in pre-processing stage of ML process flow in details.

Question-4: Discuss concept of hyperparameter tuning a ML model in brief? Explain Grid Search and Random Search method using suitable example.

Question-5: Explain Regression and Classification problem using suitable diagram and example.

Question-6: Discuss the concept of Ensemble learning in brief. Explain Bagging and Boosting method using appropriate diagram.

Question-7: What do you understand by Bias and Variance? Explain in detail. Also explain Bias-Variance tradeoff using suitable example.

Question-8: What do you understand by performance metrics/KPI? Explain in brief. Also list various KPIs used to evaluate regression and classification model.